Historical evolution of lexical concreteness re-revisited

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Exoteric languages were argued to have evolved for increased learnability due to pressures imposed by inter-group communication. One aspect relevant to learnability is lexical concreteness. Previous research has shown that English, as a globally spoken language, indeed exhibits a lexical concretization bias. We aim to replicate this result, employing, first, a different way of reconstructing lexical concreteness based on historical corpus data by integrating information about sense-level concreteness, and second, a different operationalization of concretization. Although we still find evidence for a weak concretization bias in English, we cannot confirm previous results about the interaction between concretization and frequency. Potential methodological and theoretical reasons for this discrepancy are discussed.

1. Introduction and background

The present study replicates Snefjella et al.'s investigation of lexical concretization in English (2019) to contribute to the broader discussion on the evolution of exoteric languages. It has been proposed that languages used for exoteric purposes, i.e., inter-group communication, are more regular, phonologically and morphologically simpler, and semantically more transparent than purely esoteric languages, i.e., languages used only for intra-group communication (Wray & Grace, 2007; Lupyan & Dale, 2010; cf. Trudgill, 1989, 2002; Thurston, 1989). As a language expands, it is increasingly acquired not only by child learners but also by adult learners. Consequently, the language is subject to pressures to enhance adult learnability (cf. Wray & Grace, 2007; Raviv et al. 2021) and, more generally, ease of processing in adults. This may manifest in various ways, one of which is lexical concreteness: words with concrete senses are generally easier to acquire and memorize than words with abstract senses (Hills & Adelman, 2015; cf. Reijnierse et al., 2019). Since features that are easier to process should be selected for (Hills & Adelman, 2015), while features that are harder to process should be selected against (Lupyan & Dale, 2010), we could expect the lexicon of a language that is used for exoteric purposes, such as English, to become increasingly concrete over time (see Section 5 for a discussion of opposing dynamics).

Over the past 200 years, English has gained prominence as a global *lingua franca* (Crystal, 2012; Graddol, 2006); it has thus come to be used exoterically. The pressures towards higher learnability associated with this function appear to have contributed to recent changes in its lexicon: Upon examining the diachronic development of English using the Google Ngrams corpus, COHA, and U.S. presidents' inaugural addresses, Hills and Adelman (2015) observed a concretization bias. Additionally, Hills and Adelman (2015) found a correlation between rising lexical concreteness and U.S. immigration patterns and argue that the influx of second-language learners exerted pressure on the language to become easier for adults to acquire and therefore more concrete.

Hill and Adelman's (2015) proposal was supported by Snefjella et al. (2019), who reconstructed lexical concreteness scores for English words for each decade from 1850 to 2010 based on word embeddings. Their findings indicate that, on average, the English lexicon has become increasingly concrete over time and that high-frequency words display a higher concretization rate than low-frequency ones. In our replication of their study, we opt for a sense-based approach rather than relying on reconstruction via historical word embeddings. This has the advantage of capturing differences in concreteness across the various senses of polysemous words. While Snefjella et al.'s (2019) results could be replicated using their reconstructed concreteness scores together with our way of measuring concretization, their findings could only be partially confirmed on the basis of our sense-informed scores. Although we also detected a tendency towards increasing concreteness over time, the concretization rate observed was considerably lower, and word frequency had no effect on the concretization rate.

In the discussion, we deliberate the effect of the methodological differences between the present study and Snefjella et al. (2019) and relate our findings to the competing mechanisms that may drive lexical evolution toward either increasing concreteness or abstraction over time.

2. Data and operationalization of concretization

Snefjella et al. (2019) have employed a rating-propagation algorithm adopted from Hamilton et al. (2016) to reconstruct lexical concreteness scores for all decades from 1850 to 2010. In a nutshell, this is done by inferring a word's concreteness in a given decade from a set of seed words that are semantically near in that decade. Semantic proximity is operationalized through historical word embeddings (fastText). Seed-word concreteness was taken from Brysbaert et al.'s (2014) list of crowdsourced concreteness ratings on a 5-point scale for about 40,000 English lemmas.

Apart from the observation that embedding similarity might be affected by frequency (Dubossarsky et al. 2017), one potential issue with this type of reconstruction is that embeddings do not directly encode polysemy. A word's senses might vary in their concreteness, however. For this reason, we adopt a different and, crucially, sense-informed approach, already employed by Tiessler et al. (2025) for a similar purpose (reconstruction of lexical emotion scores).[1]

We use a dataset compiled by Hu et al. (2019) consisting of decade-wise sense probabilities (inferred from the Corpus of Historical American English, COHA, through a sense-disambiguation model) and corresponding sense descriptions taken from the Oxford English Dictionary (OED). Crucially, these descriptions are written in contemporary English. All sense descriptions were tokenized, lemmatized, and PoS tagged with UDPipe. We adopt methods from lexicon-based sentiment analysis (Taboada et al., 2011) and compute, for each sense $s_{w,i}$ of a word w, concreteness $c(s_{w,i})$ as the mean of the contemporary concreteness scores of all content words (noun, verb, adj, adv) in the sense's description (as shown in its OED entry) that surface in the concreteness lexicon of Brysbaert et al. (2014).

For each decade t, we then define the word's concreteness as

$$\mathcal{C}(w,t) := \sum_{i=1}^{n} c(s_{w,i}) \cdot p(s_{w,i}, t) \tag{1}$$

where $p(s_{w,i}, t)$ is the probability $s_{w,i}$ of sense at time t, and where n is the number of senses of w. Thus, $\mathscr{C}(w,t)$, t = 1850s, 1860s,…, 2010s, defines a concreteness trajectory for all w. For each word, concretization rate r_w is then computed as the slope of a linear Gaussian (least-squares) regression model $\mathscr{C}(w,t) = r_w \cdot t + d + \epsilon$. Positive r_w indicates concretization, while negative r_w corresponds to cases of lexical abstraction or bleaching. Figure 1 exemplarily displays trajectories of strongly increasing and decreasing trajectories based on an arbitrary threshold of $r_w = \pm 0.25$ concreteness points per century. What is evident, anyway, is that there are more cases of concretization than of bleaching, which is in line with Snefjella et al. (2019).

[1] All computations were done in R (R Core Team 2025). Data and code are available at: https://osf.io/keb9r/

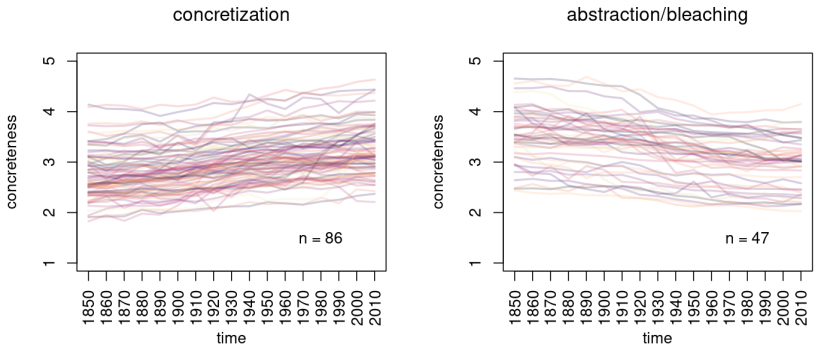

Figure 1. Trajectories of words with high (left; $r > 0.25$; top 5: *cop*, *gravity*, *abuse*, *naked*, *sore*) and low (right; $r < -0.25$: top 5: *major*, *bus*, *engine*, *margin*, *verge*) concretization rates. There are more cases of strong concretization than there are cases of strong bleaching.

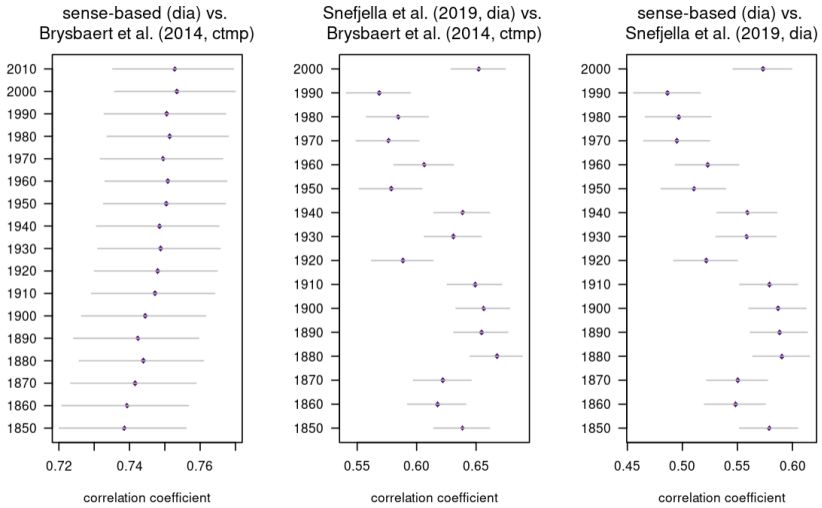

Figure 2. Correlations (Pearson's R) among concreteness scores. Left: sense-based diachronic scores vs. contemporary scores from Brysbaert et al. (2014). Middle: diachronic scores from Snefjella et al. (2019) vs. contemporary scores from Brysbaert et al. (2014). Right: period-wise correlation between sense-based scores and scores from Snefjella et al. (2019).

3. Quality assessment

Given the lack of a historical gold-standard data for lexical concreteness, we restricted ourselves to comparing our reconstructed diachronic scores $\mathscr{C}(w,t)$ against (i) scores in the contemporary lexicon by Brysbaert et al. (2014) and (ii) concreteness scores reconstructed by Snefjella et al. (2019) per decade. In addition, we tested the diachronic scores in Snefjella et al. (2019) against the contemporary ones in Brysbaert et al. (2014). For all comparisons, we straightforwardly computed Pearson's R.

We found that (i) our sense-based scores correlate strongly with those in the concreteness lexicon (Figure 2, left), and that the correlation gets weaker, albeit not significantly so, if we go back in time, which intuitively makes sense. However, the period-wise comparison of our scores with those in Snefjella et al. (2019) reveal significantly lower correlations and a somewhat erratic pattern (Figure 2, right). Indeed, comparison (ii) shows that this pattern is inherited from substantially varying correlations between diachronic ratings in Snefjella et al. (2019) and the contemporary gold-standard. This suggests that methodological differences have a crucial effect on the reconstruction of concreteness ratings.

4. Concretization bias and the impact of frequency

We wanted to test the two hypotheses that were supported by Snefjella et al.'s (2019) results: (i) words on average tend to become more concrete (rather than more abstract), and (ii) high-frequency words display a higher tendency for concretization than low-frequency words.

We first replicated Snefjella et al.'s (2019) findings based on their own data. To this end, we computed concretization rates r_w based on their concreteness ratings, and subsequently fitted a Bayesian linear model of r_w depending on frequency f_w (where f_w measures the logged average frequency in COHA across all decades) with `rstanarm` (Goodich et al. 2025). Figure 3 (left) shows that the results match those in the original study.

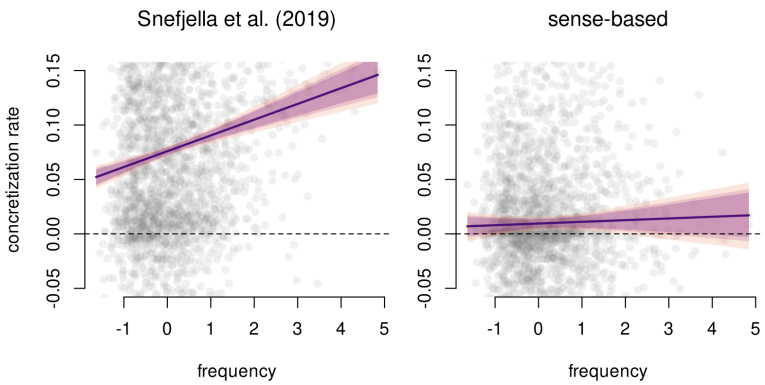

Figure 3. Bayesian linear models of concretization rate depending on frequency (log). Left: scores from Snefjella et al. (2019). Right: sense-based scores from our study. Credible intervals: 0.90, 0.95, 0.99 (from dark to light).

However, when fitting an analogous model of r_w depending on frequency f_w using our sense-based scores $\mathscr{C}(w,t)$ (Figure 3, right), the picture changes. While for most words in the mid-frequency range, a concretization bias is still retained,

which largely confirms (i) above, that bias is about one order of magnitude smaller than that derived from Snefjella et al.'s (2019) ratings. What is more striking, however, is that the positive effect of frequency on concretization rate vanishes in the sense-based setting, thus providing no support for hypothesis (ii).

5. Discussion and conclusion

Our replication of Snefjella et al.'s (2019) study lends further support to their hypothesis that English words on average tend to become more concrete in the timespan from 1850 to 2000, but casts doubt on their hypothesis that high-frequency words display a higher tendency for concretization than low-frequency words. There are several potential explanations for this finding. When working with the kinds of datasets we are dealing with here, we always have to keep in mind that they are prone to artifacts that are due to the methodological approach(es) chosen. For example, we know from Reijnierse et al. (2019) that concreteness ratings reflect the concreteness of the source sense, which is typically more concrete. This could have led to an overestimation of concreteness in Snefjella et al.'s (2019) historical scores, which in turn would imply that there could be less potential for an increase in concreteness, leading to lower overall concretization rates. The fact that hypothesis (ii) cannot be confirmed on the basis of our data perhaps also has to do with the methods we chose: High-frequency words tend to have more sense entries, which may lead to smaller concreteness scores on average, making extreme changes less likely.

Setting aside methodological considerations, there are several reasons to expect languages to undergo a decrease rather than an increase in lexical concreteness over time. One frequently observed type of lexical change is semantic bleaching, whereby a word's meaning becomes diluted, less specific, and hence more abstract over time. Closely related is the process of grammaticalization, in which a lexical item is recruited to fulfil a grammatical function and, in doing so, undergoes abstraction (Hopper & Traugott, 2003; Narrog & Heine, 2021). A decrease in concreteness may also arise through semantic widening, particularly via metaphorical extension: relatively concrete words may gain additional, more abstract senses. Indeed, concrete words are particularly susceptible to semantic diversification (Baumann & Hartmann, 2025). Even when the novel senses are not necessarily more abstract than the original sense of the source word (Winter & Srinivasan, 2022), the accumulation of multiple senses typically renders the word more abstract. Overall, opposing forces may shape lexical change, pushing it toward either greater concreteness or greater abstraction over time.

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